1. Clinical Question & Core Hypothesis

Clinical Question

From the Ontologist synthesis, TIL/immune signals and DNA-repair (MSH6) and stromal/metastasis drivers (GREM1) all bear on prognosis, yet they do not always align — i.e., some TIL-rich or MMR-deficient tumors still have poor outcomes. Why is immune infiltration sometimes ineffective? 2) Tier 1 paths used: I built on MSH6 (core DNA mismatch-repair / neoantigen → immune attractor) and GREM1 (BMP-antagonist → stromal remodeling / metastasis) because together they provide a plausible mechanistic axis that can explain when immune presence fails. 3) Tier 2 support: EV cargo (MDK, KRT17) supports a tumor-secreted program that could reinforce metastatic/stromal phenotypes; methylation markers (SDC2/TFPI2) and metabolic/renal signals (GLU, creatinine, microalbuminuria) supply orthogonal, noninvasive readouts to stratify these states; PSA/pZVR represent additional cross-cancer/proteomic signals to test generalizability. 4) Endpoint choice: Overall survival (OS) — chosen because it is the most robust, ultimate clinical outcome available in the datasets and best captures the net effect of competing tumor–immune–stromal programs. 5) Deliberate exclusions: I excluded MDK and KRT17 from the core hypothesis to keep the mechanistic claim sharp (they are placed as secondary_modifiers as noninvasive EV-readouts). I also excluded exploratory/indirect signals (HDL-C, CSF-AD biomarkers) from the main claim because their links are too distal?

Working Hypothesis

In bladder cancer, tumors that are MSH6-deficient and TIL-rich will exhibit variable overall survival depending on GREM1-driven stromal remodeling. Specifically, among MSH6-deficient, TIL-high tumors, those with high GREM1 expression/signature (indicating active BMP antagonism and stromal remodeling) will have poorer OS than those with low GREM1 expression. This implies a mechanistic uncoupling between immune cell presence and effective anti-tumor activity mediated by the stroma. The hypothesis predicts a significant interaction between MSH6 status, TIL density, and GREM1 expression in predicting OS, even after controlling for standard clinical covariates. Secondary, noninvasive readouts (EV cargo MDK, KRT17; methylation markers SDC2/TFPI2; metabolic/renal signals like GLU/creatinine/microalbuminuria; PSA/pZVR) will stratify this axis and enable monitoring of state transitions (immune-attractor state vs. immune-suppressive/stromal-dominated state). Mechanistic corollaries include: (1) GREM1-high TMEs will show reduced cytotoxic T cell activity markers and ECM/fibroblast activation signatures; (2) BMP signaling modulation will influence T cell function and tumor control in co-culture/in vivo models; (3) interventions targeting BMP/GREM1-driven stromal programs will enhance the efficacy of immune-directed therapies in GREM1-high, MSH6-deficient contexts.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MSH6, GREM1
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6 (core DNA mismatch-repair / neoantigen → immune attractor) and GREM1 (BMP-antagonist → stromal remodeling / metastasis) because together they provide a plausible mechanistic axis that can explain when immune presence fails.

GREM1: GREM1 (BMP-antagonist → stromal remodeling / metastasis) because together they provide a plausible mechanistic axis that can explain when immune presence fails.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	MDK	Supporting modifier from knowledge graph
Secondary	KRT17	Supporting modifier from knowledge graph
Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph
Secondary	GLU	Supporting modifier from knowledge graph
Secondary	urinary creatinine	Supporting modifier from knowledge graph
Secondary	microalbuminuria	Supporting modifier from knowledge graph
Secondary	PSA	Supporting modifier from knowledge graph
Secondary	pZVR	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: The interaction between the imaging feature and anchor gene predicts clinical outcome.

Prediction 2: Spatial analysis will confirm the mechanistic link at the tissue level.

Prediction 3: Functional perturbation will establish causality.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- MDK: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- KRT17: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.